

## CHANGCHUN, China

WMO: 541610

Lat: 43.901N

Lon: 125.209E

Elev: 238

StdP: 98.50

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -25.8      | -23.2 | -30.8                         | 0.2 | -23.7 | -28.6 | 0.3 | -21.3 | 9.6                   | -7.1 | 8.4 | -7.0 | 2.0                   | 220  | 0.539 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.3                    | 31.2            | 20.9 | 30.0 | 21.0 | 28.8 | 20.7 | 24.5                | 28.3 | 23.5 | 27.3 | 22.7 | 26.4 | 4.8                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.2                            | 18.5 | 26.8 | 22.2 | 17.4 | 25.9 | 21.5 | 16.6 | 25.2 | 75.6          | 28.3 | 71.9 | 27.5 | 68.8 | 26.3 | 28.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 10.3          | 8.6  | 7.4 | DB                         | -28.6 | 33.8               | 3.6 | 1.6                                                | -31.2 | 34.9       | -33.3 | 35.8       | -35.3 | 36.6       | -37.9 | 37.8 | (4) |
| (5)               |      |     | WB                         | -28.8 | 25.4               | 3.5 | 1.2                                                | -31.3 | 26.3       | -33.4 | 27.0       | -35.4 | 27.6       | -38.0 | 28.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 6.3    | -14.8 | -9.9  | -1.5  | 8.5   | 15.9  | 21.4  | 23.8  | 22.2  | 16.4  | 7.8   | -3.1  | -12.2 | (6)  |  |
|                                                                        | DBStd       | 14.30  | 4.93  | 5.82  | 5.74  | 5.22  | 4.07  | 3.26  | 2.32  | 2.66  | 3.66  | 5.00  | 6.36  | 5.41  | (7)  |  |
|                                                                        | HDD10.0     | 2960   | 769   | 556   | 357   | 88    | 3     | 0     | 0     | 2     | 102   | 395   | 688   |       | (8)  |  |
|                                                                        | HDD18.3     | 4832   | 1028  | 789   | 614   | 296   | 98    | 9     | 0     | 4     | 79    | 326   | 643   | 947   | (9)  |  |
|                                                                        | CDD10.0     | 1611   | 0     | 0     | 1     | 44    | 187   | 342   | 427   | 379   | 194   | 34    | 1     | 0     | (10) |  |
|                                                                        | CDD18.3     | 441    | 0     | 0     | 0     | 2     | 23    | 102   | 169   | 124   | 21    | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 3372   | 0     | 0     | 0     | 24    | 252   | 852   | 1266  | 827   | 146   | 7     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 889    | 0     | 0     | 0     | 2     | 61    | 273   | 351   | 186   | 16    | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.3    | 2.9   | 3.3   | 3.9   | 4.4   | 4.1   | 3.3   | 2.9   | 2.6   | 2.8   | 3.4   | 3.5   | 3.1   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 580    | 3     | 4     | 11    | 23    | 43    | 85    | 189   | 126   | 53    | 29    | 10    | 4     | (15) |  |
|                                                                        | PrecMax     | 822    | 9     | 17    | 33    | 76    | 89    | 199   | 444   | 350   | 157   | 104   | 34    | 16    | (16) |  |
|                                                                        | PrecMin     | 330    | 0     | 0     | 0     | 0     | 5     | 19    | 43    | 24    | 13    | 1     | 1     | 0     | (17) |  |
|                                                                        | PrecStd     | 106    | 2     | 4     | 10    | 17    | 23    | 41    | 89    | 78    | 33    | 24    | 7     | 4     | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 0.9   | 7.6   | 17.3  | 26.4  | 30.9  | 34.0  | 33.0  | 31.4  | 28.3  | 24.1  | 15.8  | 5.1   | (19) |  |
|                                                                        |             | MCWB   | -1.9  | 1.9   | 8.0   | 12.7  | 16.2  | 19.1  | 23.2  | 23.0  | 19.3  | 14.6  | 9.6   | 3.1   | (20) |  |
|                                                                        | 2%          | DB     | -2.3  | 4.4   | 12.9  | 22.9  | 28.3  | 31.5  | 31.1  | 30.0  | 26.8  | 20.8  | 11.9  | 0.9   | (21) |  |
|                                                                        |             | MCWB   | -4.5  | 0.1   | 5.3   | 11.5  | 15.0  | 18.3  | 22.8  | 22.8  | 18.5  | 12.9  | 6.9   | -1.3  | (22) |  |
|                                                                        | 5%          | DB     | -4.5  | 2.1   | 10.0  | 20.2  | 26.2  | 29.8  | 29.9  | 28.8  | 25.0  | 18.1  | 9.1   | -1.2  | (23) |  |
|                                                                        |             | MCWB   | -6.5  | -1.5  | 3.3   | 10.1  | 14.4  | 18.4  | 22.5  | 22.3  | 17.2  | 11.4  | 5.3   | -3.1  | (24) |  |
|                                                                        | 10%         | DB     | -6.8  | -0.2  | 7.5   | 17.9  | 24.0  | 28.0  | 28.7  | 27.4  | 23.3  | 16.0  | 6.2   | -3.8  | (25) |  |
|                                                                        |             | MCWB   | -8.5  | -3.1  | 2.0   | 8.7   | 13.4  | 18.3  | 22.2  | 21.4  | 16.2  | 10.5  | 2.9   | -5.6  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | -1.8  | 2.5   | 8.7   | 15.0  | 18.6  | 22.8  | 26.0  | 25.7  | 21.8  | 16.1  | 10.3  | 2.7   | (27) |  |
|                                                                        |             | MCDB   | 0.3   | 6.1   | 16.2  | 22.3  | 26.4  | 27.1  | 30.1  | 28.8  | 25.6  | 20.6  | 14.3  | 4.8   | (28) |  |
|                                                                        | 2%          | WB     | -4.4  | 0.4   | 6.0   | 12.5  | 16.7  | 21.3  | 24.8  | 24.3  | 20.2  | 14.2  | 7.5   | -1.0  | (29) |  |
|                                                                        |             | MCDB   | -2.5  | 3.8   | 12.2  | 21.2  | 24.9  | 27.2  | 28.7  | 28.2  | 24.5  | 19.0  | 11.1  | 0.5   | (30) |  |
|                                                                        | 5%          | WB     | -6.4  | -1.3  | 4.0   | 10.7  | 15.6  | 20.4  | 23.9  | 23.5  | 18.9  | 12.6  | 5.5   | -3.2  | (31) |  |
|                                                                        |             | MCDB   | -4.6  | 2.0   | 9.3   | 19.2  | 23.0  | 26.1  | 27.7  | 27.1  | 22.9  | 17.3  | 8.6   | -1.5  | (32) |  |
|                                                                        | 10%         | WB     | -8.4  | -3.3  | 2.2   | 9.2   | 14.4  | 19.6  | 23.1  | 22.6  | 17.6  | 11.0  | 3.1   | -5.5  | (33) |  |
|                                                                        |             | MCDB   | -6.7  | -0.2  | 7.2   | 16.7  | 22.2  | 25.0  | 26.9  | 25.9  | 22.0  | 15.4  | 6.3   | -3.8  | (34) |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 10.0  | 10.8  | 10.6  | 11.7  | 11.2  | 10.0  | 8.3   | 8.8   | 11.4  | 11.1  | 9.3   | 9.4   | (35) |  |
|                                                                        | 5% DB       | MCDBR  | 11.8  | 11.9  | 13.4  | 14.7  | 14.1  | 12.6  | 10.0  | 10.1  | 12.6  | 13.3  | 11.4  | 11.0  | (36) |  |
|                                                                        |             | MCWBR  | 10.2  | 8.8   | 8.2   | 7.7   | 6.0   | 4.8   | 4.3   | 4.7   | 5.9   | 7.4   | 8.0   | 9.7   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 11.6  | 11.6  | 12.5  | 13.0  | 11.7  | 9.3   | 8.1   | 8.0   | 10.4  | 11.8  | 10.8  | 11.0  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.328 | 0.358 | 0.436 | 0.532 | 0.599 | 0.660 | 0.623 | 0.546 | 0.474 | 0.492 | 0.399 | 0.343 | (40) |  |
|                                                                        | taud        |        | 2.072 | 2.021 | 1.897 | 1.749 | 1.664 | 1.629 | 1.741 | 1.902 | 2.001 | 1.845 | 1.990 | 2.037 | (41) |  |
|                                                                        | Ebn at Noon |        | 757   | 803   | 782   | 738   | 702   | 661   | 681   | 720   | 740   | 641   | 654   | 686   | (42) |  |
|                                                                        | Edh at Noon |        | 104   | 131   | 171   | 215   | 241   | 251   | 222   | 183   | 153   | 157   | 114   | 99    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.22  | 3.25  | 4.31  | 5.07  | 5.67  | 5.51  | 5.10  | 4.84  | 4.40  | 3.15  | 2.12  | 1.81  | (44) |  |
|                                                                        | RadStd      |        | 0.19  | 0.23  | 0.21  | 0.37  | 0.43  | 0.51  | 0.47  | 0.38  | 0.32  | 0.21  | 0.19  | 0.14  | (45) |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | -1.13  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (4 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page